



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Second Year Examination — Semester I — 2022

SCS2206 — Mathematical Methods II

(Two (2) Hours)

269



Answer ALL questions

Number of Pages = 3

Number of Questions = 4

To be completed by the candidate

Index Number

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Important Instructions to candidates:

- The medium of instruction and questions is **English**. Students should answer in the medium of English language only.
- Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
- Write your answers on the **separate answer books** provided.
- Write your index number **CLEARLY** on each page of the answer script(s).
- The paper has **4** questions on **3** pages.
- The duration of the paper is **Two (2)** hours.
- Answer **all** the questions.
- Each question carries exactly **25 marks**.
- Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.

To be completed by the examiners

1	
2	
3	
4	
Total	

1. (a). i. Consider the sequence (x_n) recursively given by $x_1 = \sqrt{2}$ and for each $n \in \mathbb{N}$, $x_{n+1} = \sqrt{2x_n}$.
Show, by induction or otherwise, that for each $n \in \mathbb{N}$, $0 < x_n < 2$ and that for each $n \in \mathbb{N}$, $x_n < x_{n+1}$.
Deduce, with reasons, that (x_n) converges. Find $\lim_{n \rightarrow \infty} x_n$.
- ii. It is given that $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$. Use this result to prove that for each $c > 1$, $\lim_{n \rightarrow \infty} \sqrt[n]{c} = 1$.
Compute $\lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{2^n} + \frac{1}{4^n} + \frac{1}{6^n}}$.

[15 marks]

- (b). Let (a_n) be a sequence of real numbers such that $a_n \geq 0$ for each $n \in \mathbb{N}$.

- i. Prove that if $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n}$ converges.
ii. Prove that if $\lim_{n \rightarrow \infty} (n^2 a_n)$ exists, then $\sum_{n=1}^{\infty} a_n$ converges.

[10 marks]

2. (a). Let P be an approximation of π such that $|\pi - P| < \frac{1}{2} \times 10^{-n}$. Show that $Q = P + \sin P$ is an approximation of π such that $|\pi - Q| < \frac{1}{2} \times 10^{-3n}$.

[5 marks]

- (b). i. The reciprocal of a number $a > 0$ (i.e. $\frac{1}{a}$) can be computed without division by the iterative formula $x_{n+1} = x_n(2 - ax_n)$, where $n \in \mathbb{N}$. Establish this relation by applying Newton's method to some suitably chosen function $f(x)$. Here, the initial value x_1 is given.
- ii. Use the iterative formula given in part (b)(i) to prove that for each $n \in \mathbb{N}$, $\text{Rel}(x_{n+1}) = [\text{Rel}(x_n)]^2$, where for each $n \in \mathbb{N}$, $\text{Rel}(x_n)$ denotes the relative error of approximating the root of $f(x) = 0$ by x_n .
Deduce that the sequence (x_n) of approximations generated by the iterative formula $x_{n+1} = x_n(2 - ax_n)$ converges to the root of $f(x) = 0$ if and only if $0 < x_1 < \frac{2}{a}$, where x_1 denotes the initial approximation.

[10 marks]

- (c). i. Does there exist a function f defined on a closed and bounded interval $[a, b]$ in $\mathbb{R}$, where $a < b$, such that $f(a) \cdot f(b) < 0$, the sequence (x_n) of approximations produced by the bisection method converges, say to x , and $f(x) \neq 0$? Justify your answer.
- ii. Using bisection method and a suitable function $f(x)$, prove that for each $\alpha \in \mathbb{Q}^c$, there exists a sequence (r_n) of rational numbers such that $\lim_{n \rightarrow \infty} r_n = \alpha$.

[10 marks]

3. (a). i. Let $x_0, x_1, \dots, x_{2022}$ be 2023 distinct real numbers and for each $i = 0, 1, \dots, 2022$, let
- $$L_i(x) = \frac{(x - x_0) \cdots (x - x_{i-1})(x - x_{i+1}) \cdots (x - x_{2022})}{(x_i - x_0) \cdots (x_i - x_{i-1})(x_i - x_{i+1}) \cdots (x_i - x_{2022})}.$$
- Prove that for each $x \in \mathbb{R}$, $L_0(x) + L_1(x) + \cdots + L_{2022}(x) = 1$.
- ii. Let $n \in \mathbb{N}$ and $a, b \in \mathbb{R}$ be such that $a < b$. Let $h = \frac{b-a}{n}$ and for each $i = 0, 1, \dots, n$, $x_i = a + ih$. Let $\psi_n(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$. Prove that for each $a \leq x \leq b$, $|\psi_n(x)| \leq n!h^{n+1}$.
- Deduce that if $p_n(x)$ is the degree n interpolating polynomial which interpolates $f(x) = e^x$ at the evenly spaced points $0 = x_0, x_1, \dots, x_n = 1$ in $[0, 1]$, then $\max_{0 \leq x \leq 1} |e^x - p_n(x)| \rightarrow 0$ as $n \rightarrow \infty$.

[5 marks]

- (b). Approximate the integral $\int_0^1 \frac{1}{1+x^3} dx$ using Simpson's Rule by dividing the interval $[0, 1]$ into 4 sub intervals of equal length. (Round the final answer to 4 decimal places.)

[10 marks]

- (c). Use Runge-Kutta method of fourth order to numerically solve the initial value problem $\frac{dy}{dx} = 4x^2 + y$, $y(0) = 1$ at $y(0.1)$ by taking $h = 0.1$. (Round the final answer to 4 decimal places.)

[10 marks]

4. (a). i. Let G be a group such that $(ab)^2 = a^2b^2$ for all $a, b \in G$. Prove that G is abelian.
- ii. Let G be a group and let $x, g \in G$. Prove that $|x| = |g^{-1}xg|$. Deduce that for each $a, b \in G$, $|ab| = |ba|$.

[15 marks]

- (b). Let G be a group of order 2022.

- i. Prove that $S = \{g \in G : g \neq g^{-1}\}$ has an even number of elements.
- ii. Deduce that G has a non trivial cyclic subgroup.

[10 marks]
